

CSL351 : COMPUTER NETWORKS
Assignment - 9

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Google Drive :

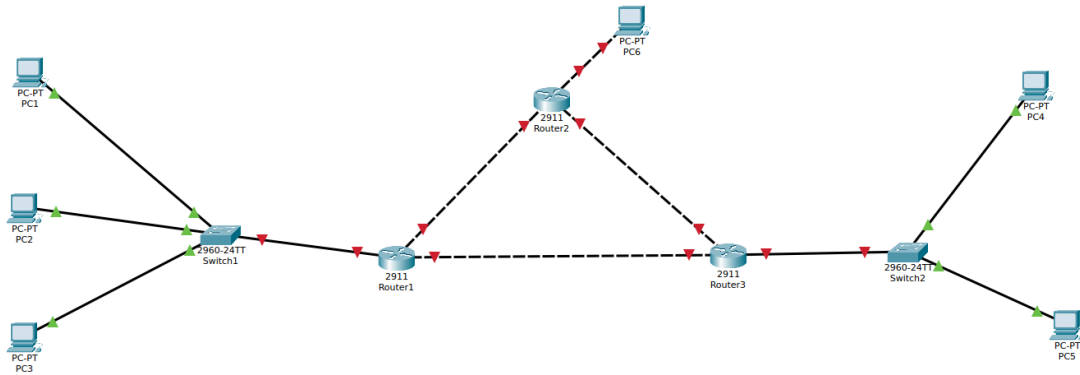
https://drive.google.com/drive/folders/1Tf33y12proMmLRiMcTHv5_TeYgByTTaz?usp=sharing

sim.webm -> the video for the packet flow in the simulation for the Part C

as9_que1.pkt -> the cisco packet tracer file which contains the configurations

Report From Next Page :

Part A :



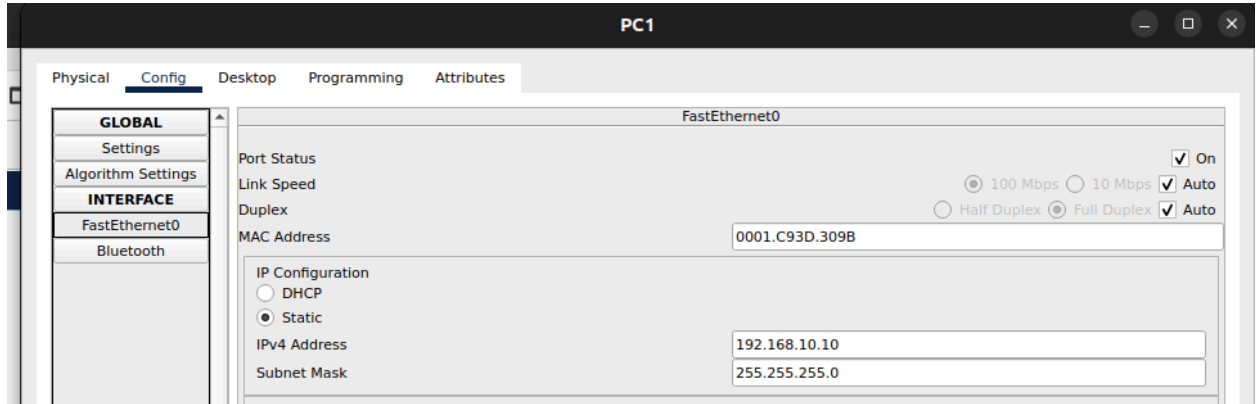
Explanation :

- I have used 3 Routers names as Router1 ,Router2, Router3 which are Router - 2911 ,then we have used two switches named as Switch1 ,Switch2 which are Switch - 2960-24TT, also we have used 6 PCs named as PC1,PC2...PC6 which are PC-PT.
- We have interconnected the routers (Router1 to Router2 , Router2 to Router3, Router3 to Router1) as per specified by the question via the Copper cross over cable using the Gigabit ethernet interface.
- The Router to Switch connection (Router1 to Switch1 ,Router3 to Switch2) is done using the copper straight through cable using the Gigabit ethernet interface.
- The Router to PC connection (Router2 to PC6) is done using the copper cross over cable using the Gigabit ethernet interface.
- The PC to Switch connection (PC1 to Switch1 ,PC2 to Switch1 ,PC3 to Switch1 ,PC4 to Switch2 ,PC5 to Switch2) is done using the copper straight through cable using the FastEthernet interface.

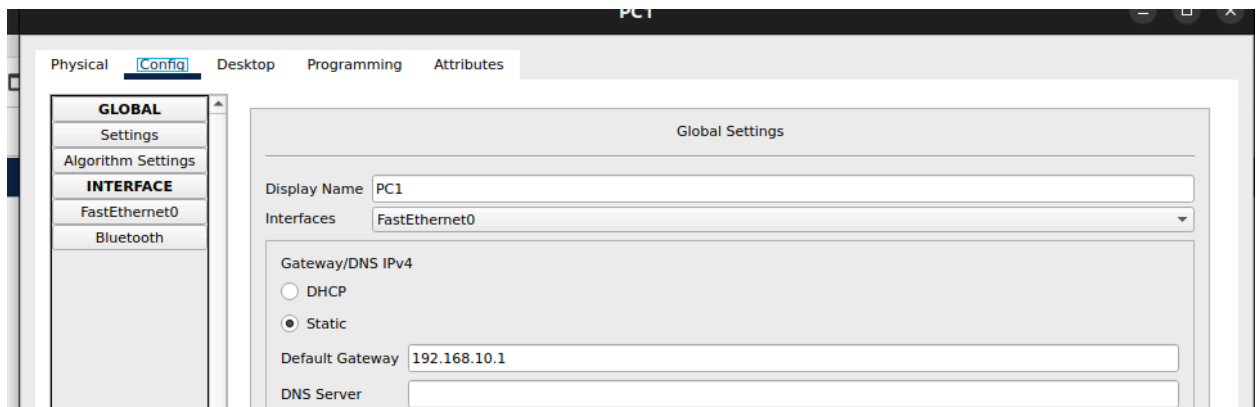
Part B :

1.) PC1 :

Assigning IP + Subnet Mask :

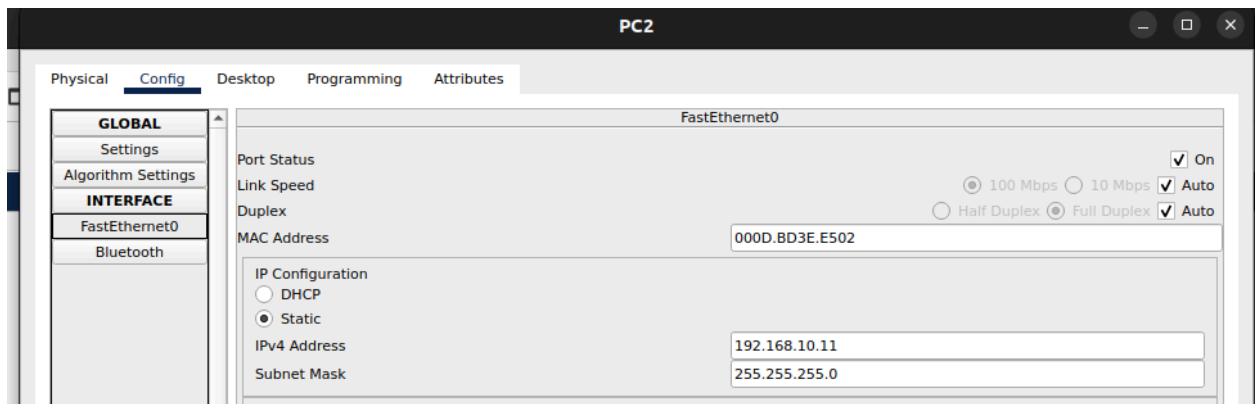


Default Gateway :

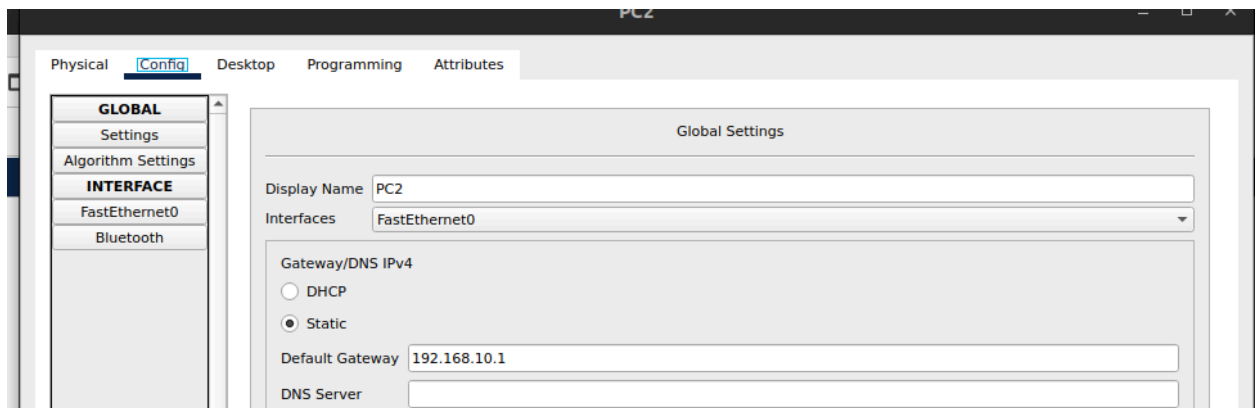


2.) PC2 :

Assigning IP + Subnet Mask :

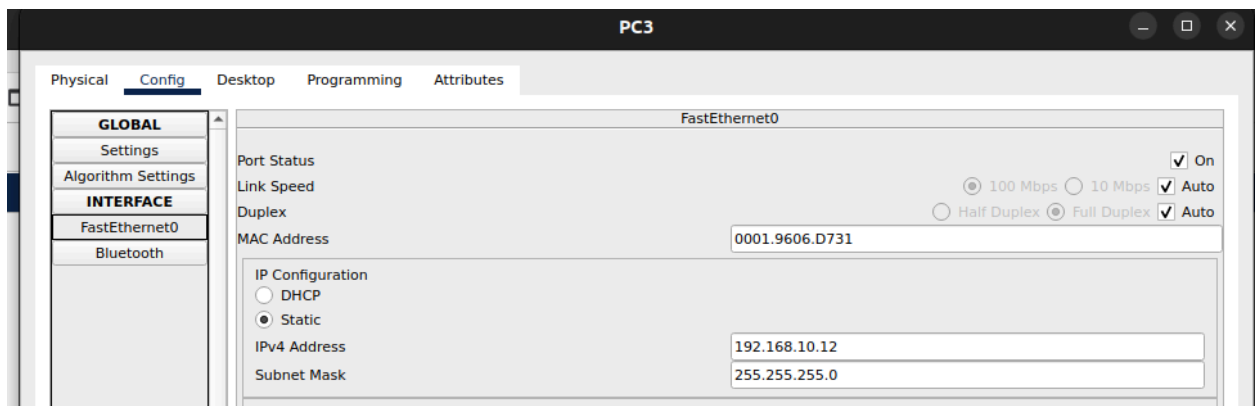


Default Gateway :

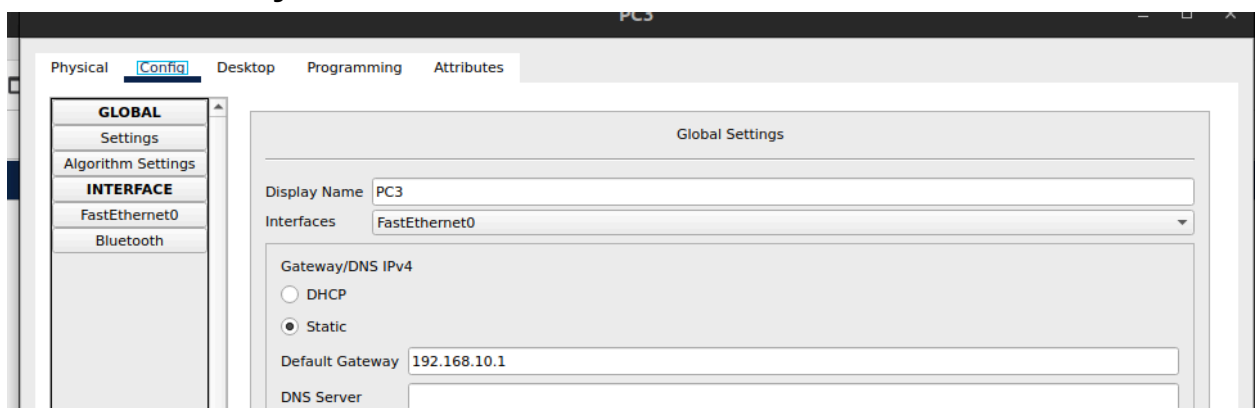


3.) PC3 :

Assigning IP + Subnet Mask :



Default Gateway :



4.) PC4 :

Assigning IP + Subnet Mask :

The screenshot shows the configuration window for PC4. The 'Config' tab is selected. In the left sidebar, under 'INTERFACE', 'FastEthernet0' is selected. The main area shows the configuration for 'FastEthernet0'. The 'Port Status' is 'On'. 'Link Speed' is set to '100 Mbps'. 'Duplex' is set to 'Full Duplex'. The 'MAC Address' is '0090.0CED.98C1'. Under 'IP Configuration', 'Static' is selected. The 'IPv4 Address' is '192.168.30.10' and the 'Subnet Mask' is '255.255.255.0'.

Default Gateway :

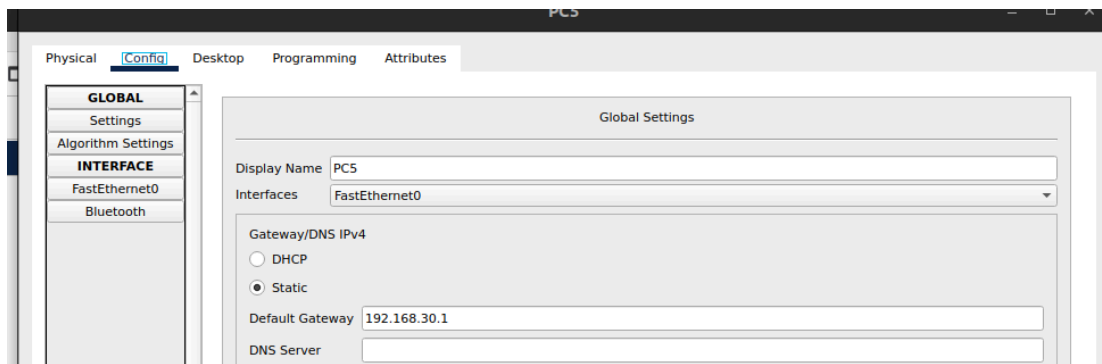
The screenshot shows the configuration window for PC4. The 'Config' tab is selected. In the left sidebar, under 'GLOBAL', 'Settings' is selected. The main area shows the 'Global Settings' for PC4. The 'Display Name' is 'PC4'. The 'Interfaces' dropdown is set to 'FastEthernet0'. Under 'Gateway/DNS IPv4', 'Static' is selected. The 'Default Gateway' is '192.168.30.1' and the 'DNS Server' is empty.

5.) PC5 :

Assigning IP + Subnet Mask :

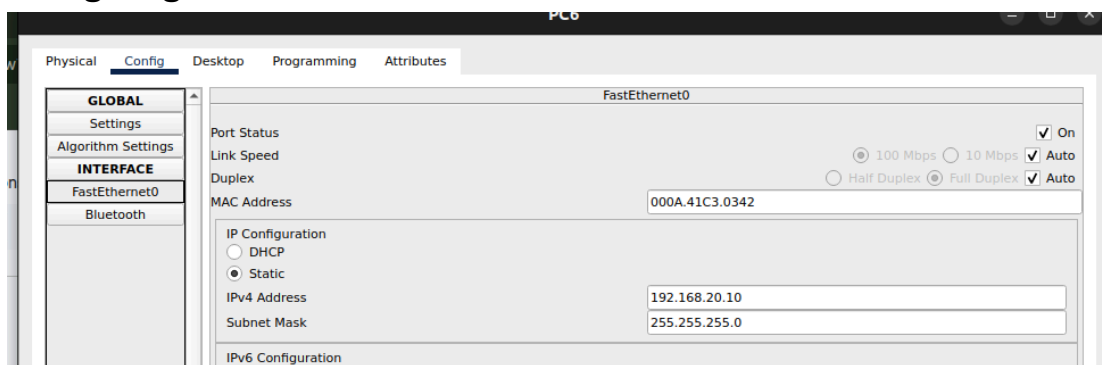
The screenshot shows the configuration window for PC5. The 'Config' tab is selected. In the left sidebar, under 'INTERFACE', 'FastEthernet0' is selected. The main area shows the configuration for 'FastEthernet0'. The 'Port Status' is 'On'. 'Link Speed' is set to '100 Mbps'. 'Duplex' is set to 'Full Duplex'. The 'MAC Address' is '0000.0C82.9BA9'. Under 'IP Configuration', 'Static' is selected. The 'IPv4 Address' is '192.168.30.11' and the 'Subnet Mask' is '255.255.255.0'.

Default Gateway :

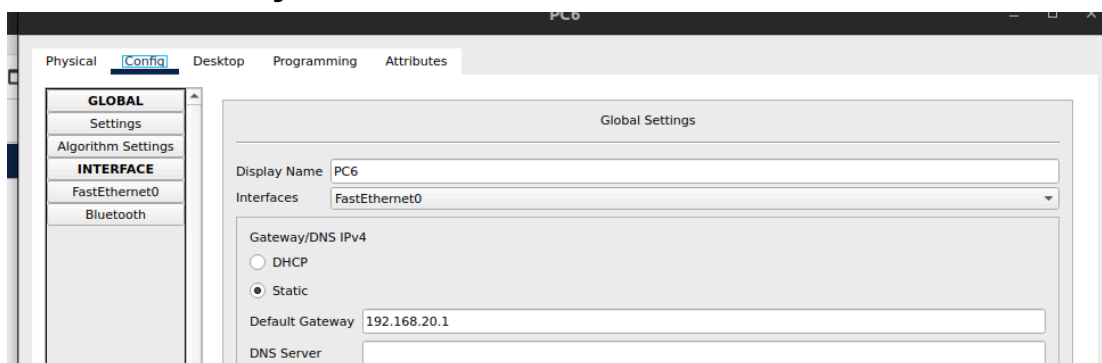


6.) PC6 :

Assigning IP + Subnet Mask :



Default Gateway :



Explanation :

- **End Device Configuration:** Each pc was assigned a ipv4 address within its respective subnet, the default gateway for each pc was set to the ip address of the first hop router interface it is connected to. (pc on switch 1 use router 1's Gig0/0 ip as their gateway). This

ensures that any traffic destined for a different network is correctly forwarded to the local router.

Configure Routers :

1. Router 1 : Assigning IP:

2911

Router1

Device Name: Router1

Device Model: 2911

Hostname: Router

Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
GigabitEthernet0/0	Up	--	192.168.10.1/24	<not set>	000C.CF78.E901
GigabitEthernet0/1	Up	--	192.168.1.1/24	<not set>	000C.CF78.E902
GigabitEthernet0/2	Up	--	192.168.3.1/24	<not set>	000C.CF78.E903
Vlan1	Down	1	<not set>	<not set>	0001.96B6.AC21

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router1

Enable Interfaces :

```
Physical Config CLI Attributes
IOS Command Line Interface

Router(config)#enable
% Incomplete command.
Router(config)#configure terminal
^
% Invalid input detected at '^' marker.

Router(config)#interface gig0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

```
Physical Config CLI Attributes
IOS Command Line Interface

Router>config t
^
% Invalid input detected at '^' marker.

Router>enable
Router#config t\
^
% Invalid input detected at '^' marker.

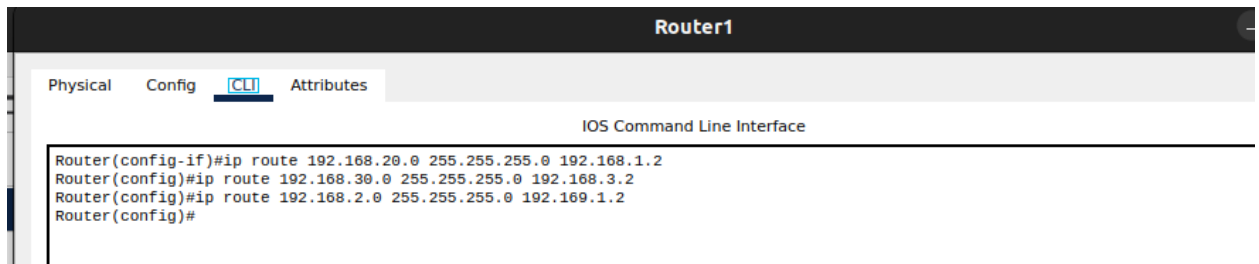
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/2
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router(config-if)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

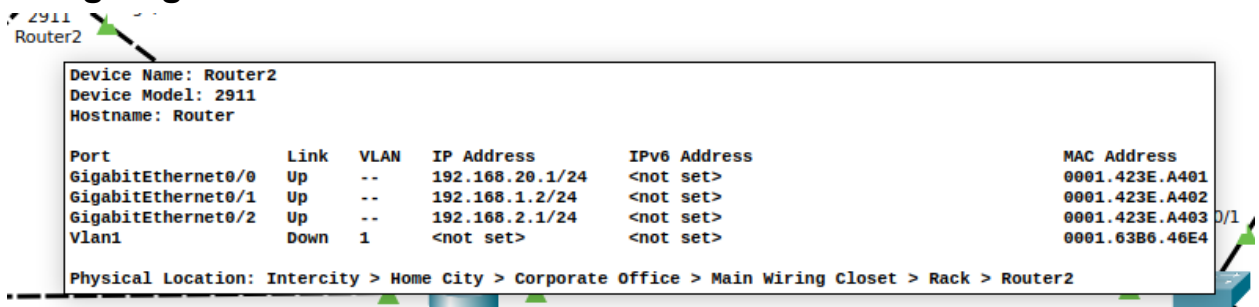
Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

Static Routing :

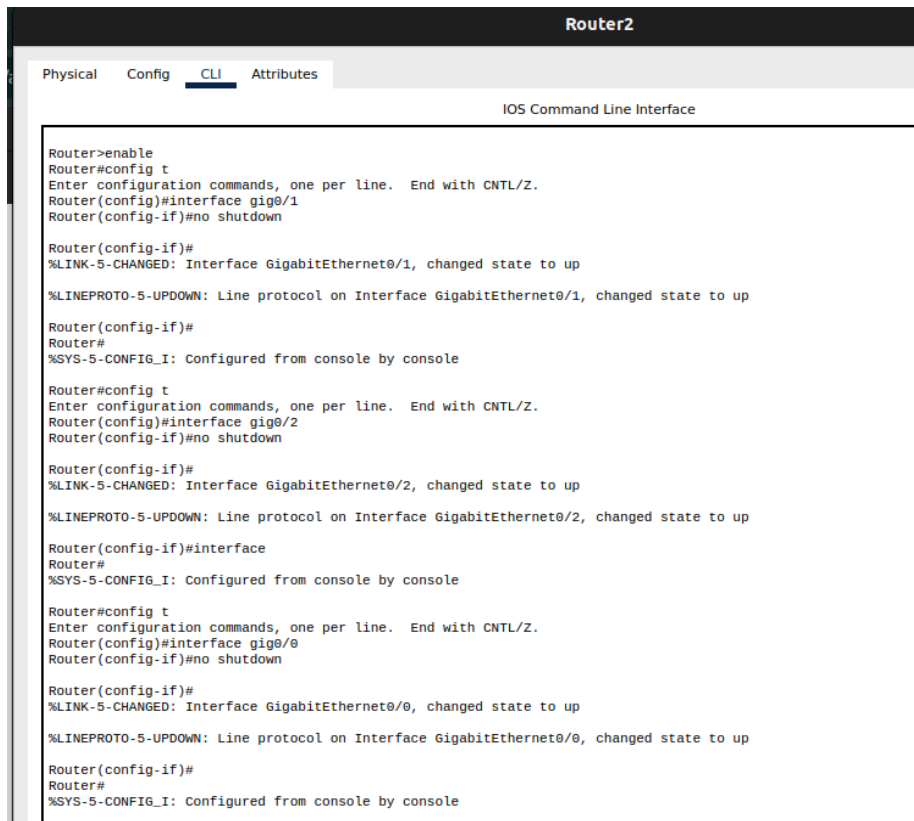


2. Router 2 :

Assigning IP :



Enabling Interfaces :



Static Routing :

Router2

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.20.0 255.255.255.0 192.168.1.2
%Invalid next hop address (it's this router)
Router(config)#ip route 192.168.10.0 255.255.255.0 192.168.1.1
Router(config)#ip route 192.168.30.0 255.255.255.0 192.168.2.2
Router(config)#ip route 192.168.3.0 255.255.255.0 192.168.1.1
Router(config)#
```

3. Router 3 : Assigning IP :

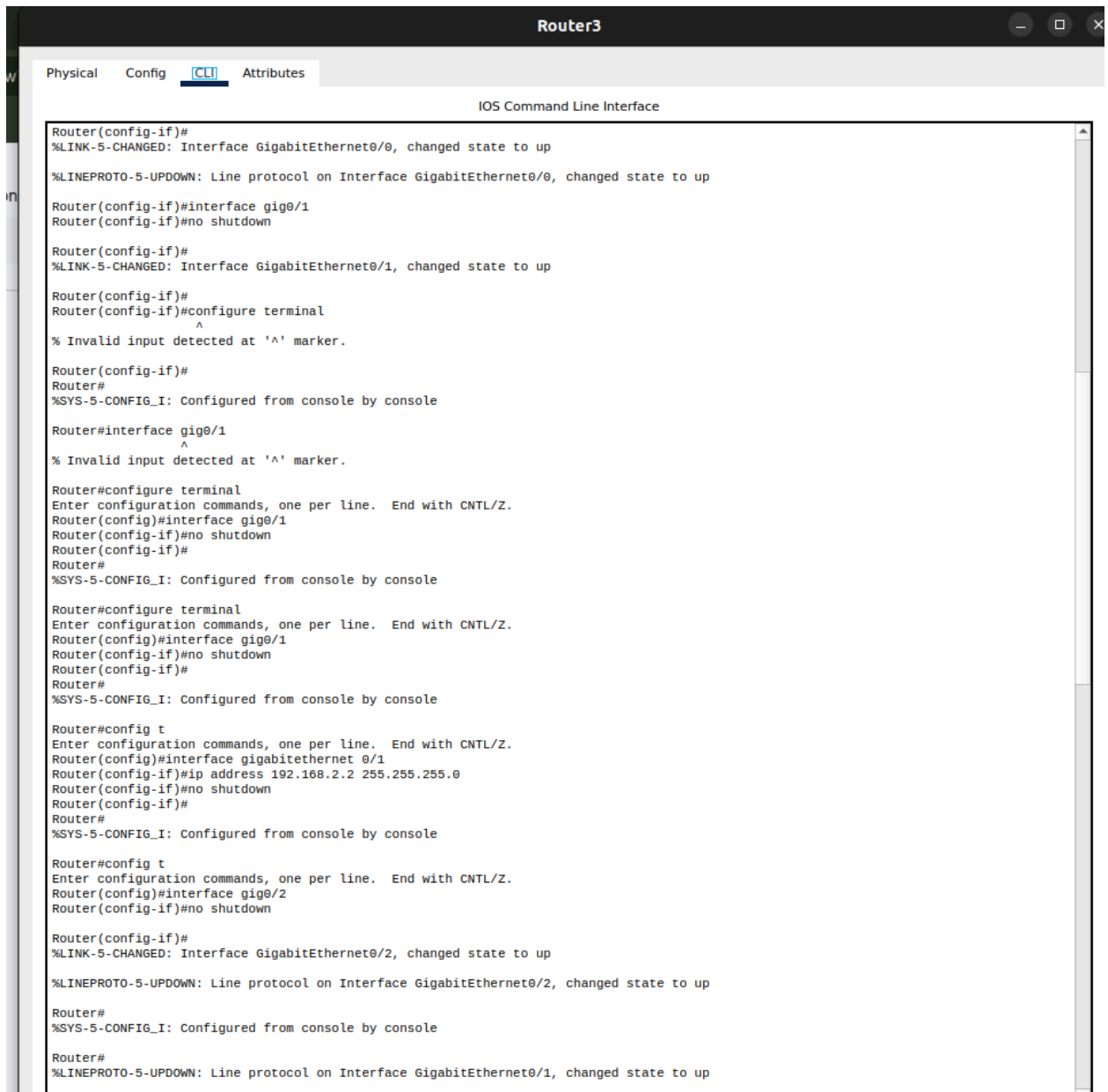
Router3

Device Name: Router3
Device Model: 2911
Hostname: Router

Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
GigabitEthernet0/0	Up	--	192.168.30.1/24	<not set>	0060.70B3.9001
GigabitEthernet0/1	Up	--	192.168.2.2/24	<not set>	0060.70B3.9002
GigabitEthernet0/2	Up	--	192.168.3.2/24	<not set>	0060.70B3.9003
Vlan1	Down	1	<not set>	<not set>	0003.E443.1EE9

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router3

Enabling Interfaces :



```
Router3
Physical Config CLI Attributes
IOS Command Line Interface

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface gig0/1
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#
Router(config-if)#configure terminal
^
% Invalid input detected at '^' marker.

Router(config-if)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#interface gig0/1
^
% Invalid input detected at '^' marker.

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/1
Router(config-if)#no shutdown
Router(config-if)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/1
Router(config-if)#no shutdown
Router(config-if)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitethernet 0/1
Router(config-if)#ip address 192.168.2.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/2
Router(config-if)#no shutdown

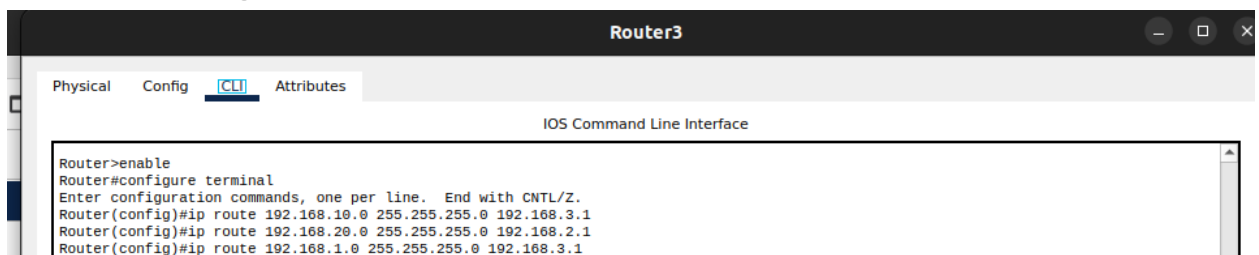
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

Static Routing :



```
Router3
Physical Config CLI Attributes
IOS Command Line Interface

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.10.0 255.255.255.0 192.168.3.1
Router(config)#ip route 192.168.20.0 255.255.255.0 192.168.2.1
Router(config)#ip route 192.168.1.0 255.255.255.0 192.168.3.1
```

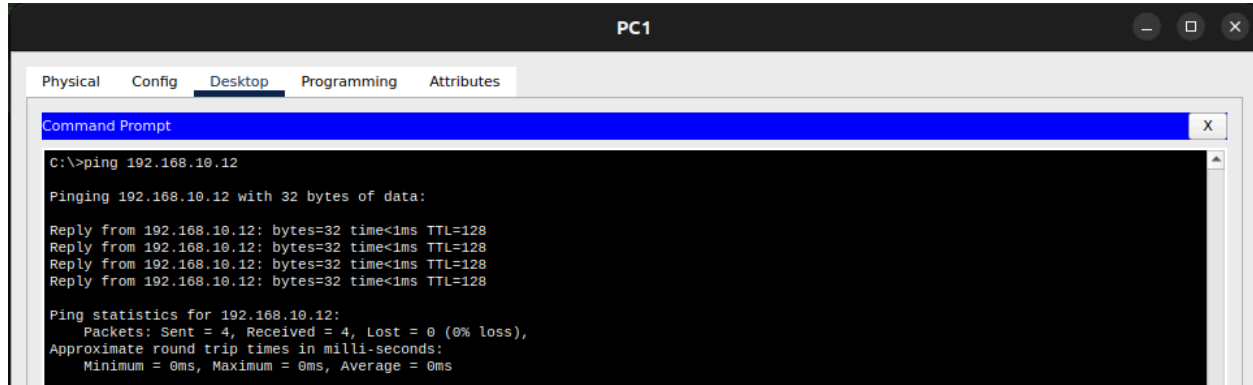
Explanation :

- **Router Interface Configuration:** The router interfaces were configured with ip addresses corresponding to the subnets they attach to and were manually activated using the no shutdown command, as router ports are disabled by default via the command prompt in the config section for each router.
- **Static Routing:** Because routers only know about networks they are directly connected to, static routing was implemented. This instructs each router on where to send packets destined for remote LANs.
Example : Router 1 was configured with static routes pointing to the 192.168.20.0 and 192.168.30.0 networks by defining the next hop ip address of the neighboring routers (Router 2 and Router 3, respectively).

Part C :

PC1 -> PC3

COMMAND USED : ping 192.168.10.12



The screenshot shows a Cisco Packet Tracer interface for PC1. The 'Desktop' tab is selected, and a 'Command Prompt' window is open. The command 'C:\>ping 192.168.10.12' has been entered. The output shows four successful replies from 192.168.10.12, each with 32 bytes, time <1ms, and TTL=128. The ping statistics indicate 4 packets sent, 4 received, 0% loss, and 0ms round trip times.

```
C:\>ping 192.168.10.12

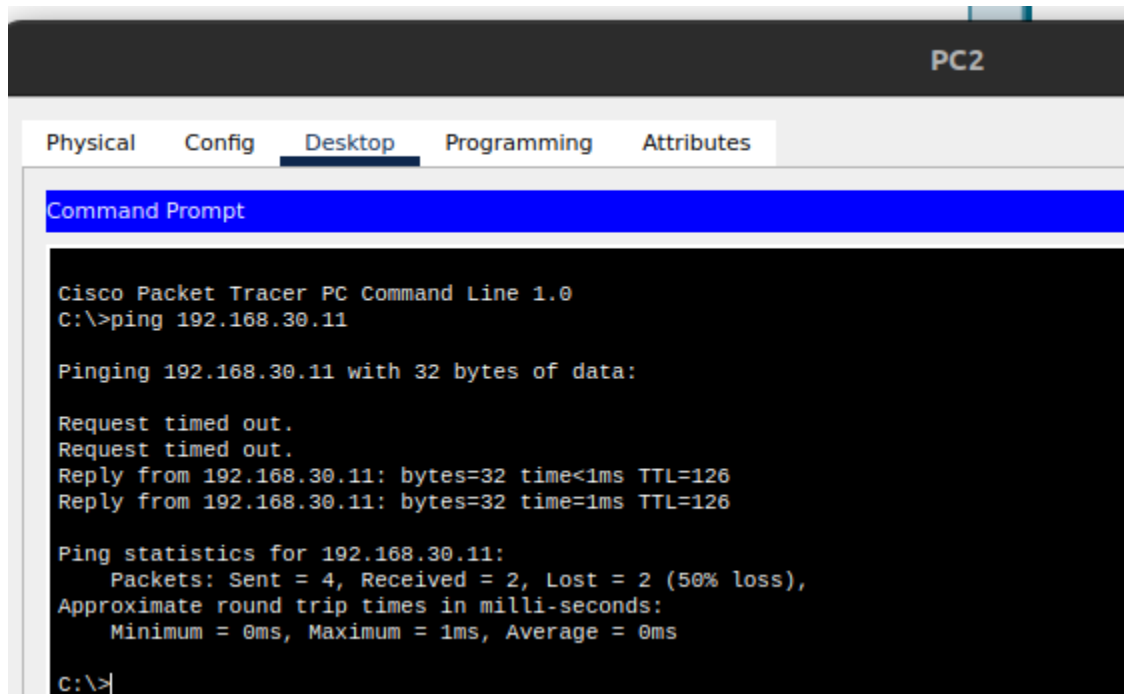
Pinging 192.168.10.12 with 32 bytes of data:

Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC2->PC5

COMMAND USED : ping 192.168.30.11



The screenshot shows a Cisco Packet Tracer interface for PC2. The 'Desktop' tab is selected, and a 'Command Prompt' window is open. The command 'C:\>ping 192.168.30.11' has been entered. The output shows two 'Request timed out' messages followed by two successful replies from 192.168.30.11, each with 32 bytes, time=1ms, and TTL=126. The ping statistics indicate 4 packets sent, 2 received, 2 lost (50% loss), and 0ms round trip times.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.11

Pinging 192.168.30.11 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.30.11: bytes=32 time=1ms TTL=126
Reply from 192.168.30.11: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.30.11:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

PC6->PC4

COMMAND USED : ping 192.168.30.10

```
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.30.10: bytes=32 time<1ms TTL=126
Reply from 192.168.30.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Explanation :

- **Local Connectivity (PC1 to PC3):**

This ping was successful with 0% packet loss. Because both pc are on the same switch and subnet (192.168.10.0/24), this verifies that the local switch 1 is functioning and pc are correctly addressed and can transmit packets, without the need of router.

- **Cross-Network with ARP Resolution (PC2 to PC5):**

This ping successfully traversed from the 192.168.10.0 network through router 1 to router 3 and down to the 192.168.30.0 network.

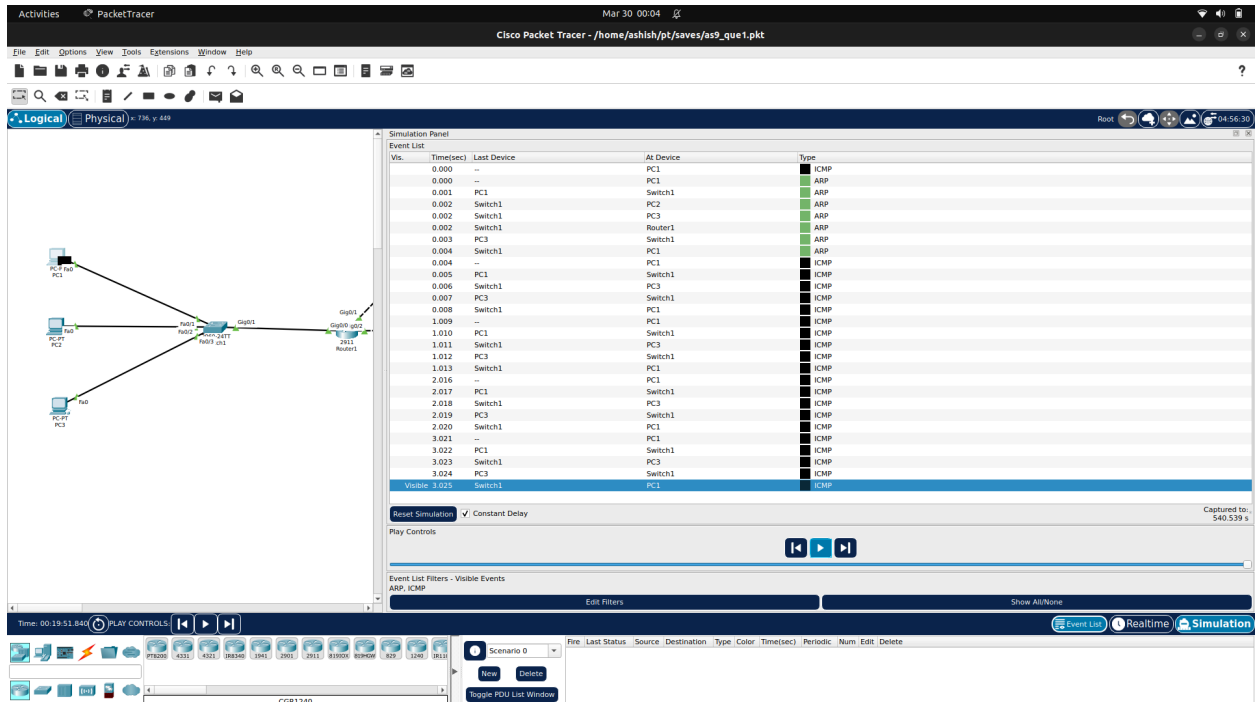
This verifies that the static routes configured on R1 and R3 are accurate. The first few request packets timed out resulting in a 50% packet loss. The timeout occurred because the routers had to perform an ARP (Address Resolution Protocol) broadcast to discover the MAC addresses associated with the target IP addresses along the new route. Once the ARP tables were updated, the subsequent ping replies were successful

- **Cross-Network with ARP Resolution (PC6 to PC4):**

When pinging from PC6 to PC4 the first few request packets timed out resulting in a 50% packet loss. The timeout occurred because the routers had to perform an ARP (Address Resolution Protocol) broadcast to discover the MAC addresses associated with the target IP addresses along the new route. After the ARP table was populated the ping replies were successful and any new ping request now from PC6 to PC4 will result in 0% loss.

Explanation for the Simulation Flow :

1. PC1->PC3



- PC1 generates the first ICMP Request meant for PC3. Since, PC1 doesn't know PC3's MAC address so it places the ICMP packet on hold. PC1 sends an ARP Request to Switch 1. Because Switch 1 doesn't know where PC3 is yet, it broadcasts this ARP request to all other connected devices on that network (PC2, PC3, and Router 1).
- PC2 and Router 1 ignore the request because the IP doesn't match theirs. PC3 recognizes its IP address and sends an ARP Reply back to Switch 1, which forwards it directly to PC1.
- Since now PC1 has PC3's MAC address, it releases the held ICMP Request. It travels smoothly from PC1 -> Switch 1 -> PC3. PC3 receives the request and immediately generates an ICMP Reply, which travels back from PC3 -> Switch 1 -> PC1.

- The first ping is successful. Because this is a fast LAN network and no router hopping is involved, the original ICMP packet does not time out.
- For the remaining pings, PC1 and PC3 have each other's MAC addresses saved in their ARP caches, and Switch 1 has the ports they are connected to. No more broadcasts are needed.
- PC1 generates the second ICMP ping. It goes straight through the switch (PC1 -> Switch 1 -> PC3) and the reply returns instantly (PC3 -> Switch 1 -> PC1). Similarly packet 3, 4 request and replies goes and comes back to PC1 via PC3 and the Switch1. All 4 pings succeed with 0% packet loss.

2. PC2->PC5

The screenshot shows the Cisco Packet Tracer interface with a network topology and an Event List panel. The network topology includes PC1, PC2, PC3, PC4, PC5, Switch1, Switch2, Router1, and Router3. The Event List panel shows a series of ICMP and ARP events. The simulation is running at 00:10:41.31.

Time(sec)	Last Device	At Device	Type
6.007	Router1	Router3	ICMP
6.007	--	Router3	ARP
6.008	Router3	Switch2	ARP
6.009	Switch2	PC4	ARP
6.009	Switch2	PC5	ARP
6.010	PC5	Switch2	ARP
6.011	Switch2	Router3	ARP
12.009	--	PC2	ICMP
12.010	PC2	Switch1	ICMP
12.011	Switch1	Router1	ICMP
12.012	Router1	Router3	ICMP
12.013	Router3	Switch2	ICMP
12.014	Switch2	PC5	ICMP
12.015	PC5	Switch2	ICMP
12.016	Switch2	Router3	ICMP
12.017	Router3	Router1	ICMP
12.018	Router1	Switch1	ICMP
12.019	Switch1	PC2	ICMP
13.023	--	PC2	ICMP
13.024	PC2	Switch1	ICMP
13.025	Switch1	Router1	ICMP
13.026	Router1	Router3	ICMP
13.027	Router3	Switch2	ICMP
13.028	Switch2	PC5	ICMP
13.029	PC5	Switch2	ICMP
13.030	Switch2	Router3	ICMP
13.031	Router3	Router1	ICMP
13.032	Router1	Switch1	ICMP
13.032	Switch1	PC2	ICMP

- PC2 generates the first ICMP Request meant for PC5. However PC2 doesn't know the MAC address of its default gateway (Router 1), so it places the ICMP packet on hold and sends out an ARP Request to Switch 1, which broadcasts it to all connected devices. Router 1

replies back to PC2 through Switch 1. Now PC2 can send the ICMP packet.

- The ICMP packet is successfully forwarded from PC2 through Switch 1 to Router 1. Router 1 looks at its routing table and sees the packet needs to go to Router 3 next. But Router 1 doesn't know Router 3's MAC address. It puts the ICMP packet on hold and sends an ARP request to Router 3. Router 3 replies. By the time Router 1 learns Router 3's MAC address, the original ICMP ping packet has timed out and is dropped. This causes our first "Request timed out".
- PC2 tries again as the path from PC2 to Router 3 is known, but Router 3 doesn't know exactly where PC5 is on its local network. PC2 generates the second ICMP ping. Because the MAC addresses for Router 1 and Router 3 are now saved in their ARP caches, the packet travels from PC2 -> Switch 1 -> Router 1 -> Router 3 without stopping. Router 3 receives the packet and it needs to send it to PC5 but it doesn't know PC5's MAC address.
- Router 3 sends an ARP broadcast down to Switch 2. Switch 2 broadcasts it to both PC4 and PC5. PC4 ignores it, but PC5 replies back to Router 3 with its MAC address. Again, while waiting for this ARP process to finish on the destination network, the second ICMP ping packet times out and is dropped. This causes our second "Request timed out".
- By the third attempt, every device along the path has the required MAC addresses saved in their ARP tables. PC2 generates the third ICMP ping. It goes straight through the network: PC2 -> Switch 1 -> Router 1 -> Router 3 -> Switch 2 -> PC5.
- PC5 receives the Request and immediately generates an ICMP Reply. The reply travels all the way back along the known path: PC5 -> Switch 2 -> Router 3 -> Router 1 -> Switch 1 -> PC2.
- The ping is successful. The fourth ping onwards travels to the destination without any timeouts like that of packet3 as the ARP tables are fully populated.

3. PC6->PC4

The screenshot shows the Cisco Packet Tracer interface. On the left, a network topology is visible with several devices connected. On the right, the 'Simulation Panel' displays an 'Event List' table. The table has columns for 'Time(sec)', 'Last Device', 'At Device', and 'Type'. The events listed are:

Time(sec)	Last Device	At Device	Type
0.000	--	PC6	ICMP
0.000	--	PC6	ARP
0.001	PC6	Router2	ARP
0.002	Router2	PC6	ARP
0.002	--	PC6	ICMP
0.003	PC6	Router2	ICMP
0.003	--	Router2	ARP
0.004	Router2	Router3	ARP
0.005	Router3	Router2	ARP
6.005	--	PC6	ICMP
6.006	PC6	Router2	ICMP
6.007	Router2	Router3	ICMP
6.007	--	Router3	ARP
6.008	Router3	Switch2	ARP
6.009	Switch2	PC4	ARP
6.009	Switch2	PC5	ARP
6.010	PC4	Switch2	ARP
6.011	--	Switch2	ARP
12.011	--	PC6	ICMP
12.012	PC6	Router2	ICMP
12.013	Router2	Router3	ICMP
12.014	Router3	Switch2	ICMP
12.015	Switch2	PC4	ICMP
12.016	PC4	Switch2	ICMP
12.017	Switch2	Router3	ICMP
12.018	Router3	Router2	ICMP
12.019	Router2	PC6	ICMP
13.020	--	PC6	ICMP
13.021	PC6	Router2	ICMP

- PC6 generates the first ICMP Request meant for PC4. However since PC6 does not have the MAC address of its default gateway(Router 2), so it places the ICMP packet on hold and sends an ARP Request to find Router 2's MAC address. Now PC6 can send the ICMP packet to Router 2.
- The ICMP packet is successfully forwarded from PC6 to Router 2. Router 2 looks at its routing table and sees the packet needs to go to Router 3 next because of the static routing we did before. But Router 2 doesn't know Router 3's MAC address. It puts the ICMP packet on hold and sends an ARP request to Router 3. This delay leads to the first "Request timed out".
- PC6 generates the second ICMP ping. Since the MAC addresses for Router 2 and Router 3 are now saved in their ARP caches, the packet travels smoothly from PC6 -> Router 2 -> Router 3 without stopping. Router 3 receives the packet from the network and it needs to send it to PC4 but it doesn't know PC4's MAC address. Router 3 sends an ARP broadcast down to Switch 2. Switch 2 broadcasts it to both PC4 and PC5. PC5 ignores it but PC4 replies back to Router 3.

with its MAC address. By the third attempt, every device along the path has the required MAC addresses saved in their ARP tables.

- PC6 generates the third ICMP ping. It goes straight through the network: PC6 -> Router 2 -> Router 3 -> Switch 2 -> PC4. PC4 receives the Request and immediately generates an ICMP Reply. The reply travels all the way back along the known path: PC4 -> Switch 2 -> Router 3 -> Router 2 -> PC6. The ping is successful as seen in the command prompt.
- The fourth ping onwards will follow the exact same path as packet 3, because the ARP tables are fully populated.